

Business Requirements Document (BRD)

Project Title: Learning Platform with Smart Study Planner (LMS) for MP Online

1. Introduction

Purpose:

The Purpose of this document is to define the business requirements for the Learning Platform with Smart Study Planner (LMS). This system will help users to manage courses, plan study schedules, track progress, and improve productivity through analytics and reminders.

Scope:

- Allow users to manage courses and study tasks.
- Provide a smart daily study planner.
- Track user progress and performance.
- Send deadline reminders and notifications.
- Offer analytics via dashboard.

Objectives:

- Improve Student's productivity and time management.
- Provide personalized study planning.
- Enable efficient tracking of learning progress.
- Reduce missed deadlines through reminders.
- Offer insights through analytics.

2. Stakeholders

- **Student / Users:** Primary Users of the platform.
- **Project Developer:** Design, Development and Maintenance.
- **Faculty / Evaluators:** Review and access project.
- **Organization:** MP Online / Service Provider
- **Admin:** System Management (Optional)

3. Overall description

Product Perspective:

The system is a standalone web-based LMS with integrated planning and analytics with reminders & notification features.

Product Features:

- User Authentication (Login/Signup)
- Course Management
- Task/Assignment Management
- Smart Study Planner
- Progress Tracking
- Notifications & Reminders
- Dashboard with Analytics

User Classes:

- Student/User: Uses platform to manage studies
- Admin (optional): Manages system data

4. Functional Requirements

User Authentication:

- Users must be able to register and login
- Secure authentication using JWT
- Password encryption

Course Management:

- Add, edit, delete courses
- View list of enrolled courses
- Assign deadlines to courses

Task Management:

- Create study tasks/assignments
- Assign priority levels (High, Medium, Low)
- Set deadlines
- Mark tasks as completed

Smart Study Planner:

- Automatically generate daily/weekly plans
- Allow manual adjustments
- Suggest study schedules based on:
 - Deadlines
 - Task priority
 - Available time

Progress Tracking:

- Track completed vs pending tasks
- Display completion percentage
- Show course-wise progress

Notifications & Reminders:

- Deadline reminders
- Daily study plan alerts
- Notifications via UI (and optional email)

Dashboard & Analytics:

- Visual charts (progress, time spent, task completion)
- Performance insights
- Weekly/monthly reports

5. Non-Functional Requirements

Performance:

- Fast response time (<2 seconds)
- Handle multiple users concurrently

Security:

- Secure authentication (JWT/OAuth)
- Password hashing (bcrypt)
- Data protection and validation

Scalability:

- Modular backend (Fast API/Flask)
- Scalable database design

Usability:

- Clean and intuitive UI
- Responsive design (mobile-friendly)

Reliability:

- System uptime $\geq 99\%$
- Error handling and logging

Maintainability:

- Clean, modular code
- API documentation (Swagger)

6. Technology Stack

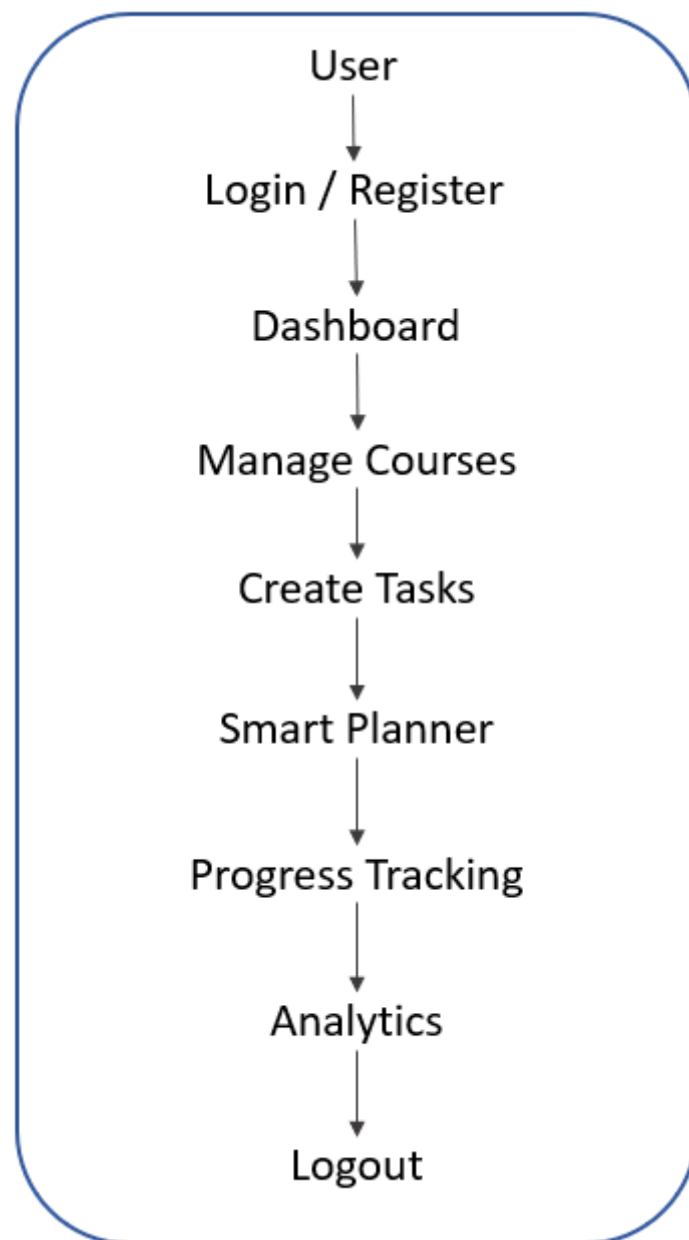
- **Frontend:** HTML, CSS, React JS, JavaScript, Tailwind CSS
- **Backend:** Fast API or Flask, RESTful APIs
- **Database:** PostgreSQL, MySQL
- **Authentication:** JWT (JSON, Web Tokens), OAuth (for Google login)
- **Tools & Libraries:** Axios (API Calls), Chart.js or Recharts (Analytics), Context API
- **Deployment:**
 - Frontend: Vercel or Netlify
 - Backend: Railway, AWS, Render

7. Business Process Flow

1. User registration or logs into the system & system validates credentials.
2. On success, user redirect to dashboard where user view pending / upcoming tasks, progress summary etc.
3. User can manage courses like add new courses, edit course details, delete course etc.
4. User then create tasks under courses. They also assigns: Priority (high, medium, low) and Deadlines.
5. Smart Study Planner:
 - System automatically: Analyzes deadlines and priorities, Generates daily/weekly study plan.
 - User can modify the plan.
6. Progress Tracking is done where system tracks completed tasks, pending tasks. It also calculates completion percentage.

7. Notifications and reminder where system sends daily study reminders and deadline alert.
8. Analytics & Dashboard: Here, system displays graphs of progress, performance insights, study trends.
9. User logs out securely and session ends.

Process Flow diagram:



8. Risk & Mitigation

Risk	Mitigation
Data loss	Use Backups
Security issues	Implement encryption
Performance issues	Optimize APIs

9. Assumption

- Users have basic internet access
- System will initially support single-user accounts
- Notifications are primarily in-app

10. Future Enhancement

- AI-based smart recommendations
- Mobile application
- Collaboration features (group study)
- Multi-language support
- Integration with online courses (Coursera, etc.)

11. Success Criteria

- Users can manage courses and tasks easily
- Study planner generates useful schedules
- Dashboard provides meaningful insights
- System performs efficiently without errors

12. Timeline

No. of Phases	Time Duration
Requirement Analysis	1 Week
Development	3 Week
Testing	1 Week
Deployment	1 Week
Total – 6 Week	